



CMM 2025 Integration Bee Qualification Test Solutions

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Problem 1. $\int_0^\infty x^7 \exp(-x^2) \, dx$

Proposed by Sam Murray

Solution: $\boxed{3}$

We let $u = x^2$ (and $du = 2x \, dx$) to obtain the following.

$$\int_0^\infty x^6 \exp(-x^2) \cdot x \, dx = \int_0^\infty \frac{u^3 \cdot \exp(-u)}{2} \, du$$

We now do tabular integration as follows, noting that $\exp(-\infty) \rightarrow 0$ and decays the polynomial terms.

$$\begin{aligned} \int_0^\infty \frac{u^3 \cdot \exp(-u)}{2} \, du &= \frac{1}{2} \cdot [\exp(-u) \cdot (-u^3 - 3u^2 - 6u - 6)]_0^\infty \\ &= \frac{1}{2} \cdot [\exp(-u) \cdot (u^3 + 3u^2 + 6u + 6)]_0^\infty \\ &= \frac{1}{2} \cdot [6 \cdot \exp(-0)] = \boxed{3} \end{aligned}$$

Problem 2. $\int_0^\pi \sin^5(x) \, dx$

Proposed by Sam Murray

Solution: $\boxed{\frac{8}{15}}$

We split $\sin^5(x)$ into a product to utilize the Pythagorean identity and utilize a u -substitution.

$$\begin{aligned} \int_0^\pi \sin^4(x) \cdot \sin(x) \, dx &= \int_0^\pi (1 - \cos^2(x))^2 \cdot \sin(x) \, dx \\ &= \int_1^{-1} -(1 - u^2)^2 \, du && (u = \cos(x)) \\ &= 2 \cdot \int_0^1 (1 - 2u^2 + u^4) \, du && (\text{even integrand}) \\ &= 2 \cdot \left[u - \frac{2u^3}{3} + \frac{u^5}{5} \right]_0^1 \\ &= 2 \cdot \left(1 - \frac{2}{3} + \frac{1}{5} \right) = \frac{2(15 - 10 + 3)}{15} = \boxed{\frac{16}{15}} \end{aligned}$$

Problem 3. $\int_{-3}^3 \frac{x^4}{1 + e^{-x}} \, dx$

Proposed by Sam Murray



Solution: $\boxed{\frac{243}{5}}$

Recall King's rule as follows.

$$\int_a^b f(x) \, dx = \int_a^b f(a+b-x) \, dx$$

Since $a = 3, b = -3$, we have that $f(a+b-x) = f(-x)$. Adding these two versions of the integral (labeled $\mathcal{J}$), we obtain the following.

$$\begin{aligned} 2\mathcal{J} &= \int_{-3}^3 \frac{x^4}{1+e^{-x}} \, dx + \int_{-3}^3 \frac{x^4}{1+e^x} \, dx \\ &= \int_{-3}^3 \frac{x^4 e^x}{e^x + 1} \, dx + \int_{-3}^3 \frac{x^4}{1+e^x} \, dx \\ &= \int_{-3}^3 \frac{x^4(e^x + 1)}{e^x + 1} \, dx \\ &= \int_{-3}^3 x^4 \, dx = \left[\frac{x^5}{5} \right]_{-3}^3 = \frac{243}{5} \cdot 2 \\ \mathcal{J} &= \boxed{\frac{243}{5}} \end{aligned}$$

Problem 4. $\int_0^1 (\sin(\log x) + \cos(\log x)) \, dx$

Proposed by Sam Murray

Solution: $\boxed{0}$

One can let $x = e^u$ (and $dx = e^u \, du$), and then perform integration by parts on the former term to simplify.

$$\begin{aligned} \int_0^1 (\sin(\log x) + \cos(\log x)) \, dx &= \int_{-\infty}^0 (\sin(u) + \cos(u)) \cdot e^u \, du \\ &= \int_{-\infty}^0 e^u \sin(u) \, du + \int_{-\infty}^0 e^u \cos(u) \, du \\ &= \left[e^u \sin(u) \Big|_{-\infty}^0 - \int_{-\infty}^0 e^u \cos(u) \, du \right] + \int_{-\infty}^0 e^u \cos(u) \, du \\ &= e^0 \sin(0) = \boxed{0} \end{aligned}$$

Problem 5. $\int_{-1}^1 \frac{\cos(x)}{\arctan(x)} \, dx$

Proposed by Sam Murray

Solution: $\boxed{0}$

The integrand is odd due to the presence of $\arctan(x)$. Since this is integrated across symmetric bounds, this integral immediately evaluates to $\boxed{0}$.



Problem 6. $\int_{\frac{\pi}{2}}^{\pi} \sqrt{1 + \cos(x)} \, dx$

Proposed by Sam Murray

Solution: $\boxed{2\sqrt{2} - 2}$

We can utilize the half-angle relation and integrate without the presence of the square root.

$$\begin{aligned} \int_{\frac{\pi}{2}}^{\pi} \sqrt{1 + \cos(x)} \, dx &= \int_{\frac{\pi}{2}}^{\pi} \sqrt{2 \cos^2\left(\frac{x}{2}\right)} \, dx \\ &= \sqrt{2} \cdot \int_{\frac{\pi}{2}}^{\pi} \left| \cos\left(\frac{x}{2}\right) \right| \, dx \\ &= \sqrt{2} \cdot \int_{\frac{\pi}{2}}^{\pi} \cos\left(\frac{x}{2}\right) \, dx && (\cos(x) \geq 0 \text{ for } x \in [0, \pi/2]) \\ &= 2\sqrt{2} \cdot \sin\left(\frac{x}{2}\right) \Big|_{\frac{\pi}{2}}^{\pi} \\ &= 2\sqrt{2} \cdot \sin\left(\frac{\pi}{2}\right) - 2\sqrt{2} \cdot \sin\left(\frac{\pi}{4}\right) = \boxed{2 - \sqrt{2}} \end{aligned}$$

Problem 7. $\int_{-\infty}^0 e^{-\sqrt[6]{-x}} \, dx$

Proposed by Sam Murray

Solution: $\boxed{720}$

We do a u -substitution of $x = -u^6$ (and $dx = -6u^5 \, du$) to reduce the complexity of the exponent as follows, and follow with tabular integration to vanish the polynomial portion of the integrand.

$$\begin{aligned} \int_{-\infty}^0 e^{-\sqrt[6]{-x}} \, dx &= \int_{\infty}^0 e^{-|u|} (-6u^5 \, du) \\ &= \int_0^{\infty} 6u^5 \cdot e^{-u} \, du \\ &= [\exp(-u) \cdot (-6u^5 - 30u^4 - 120u^3 - 360u^2 - 720u - 720)]_0^{\infty} \\ &= [\exp(-u) \cdot (6u^5 + 30u^4 + 120u^3 + 360u^2 + 720u + 720)]_0^{\infty} \\ &= [720 \cdot \exp(-0)] = \boxed{720} \end{aligned}$$

Problem 8. $\int_1^{\infty} \frac{1}{2025^{\lfloor x \rfloor}} \, dx$

Proposed by Sam Murray

Solution: $\boxed{\frac{1}{2024}}$



We can re-write the integral as an infinite summation based on each interval it is on as follows.

$$\begin{aligned}
 \int_1^{\infty} \frac{1}{2025^{\lfloor x \rfloor}} dx &= \sum_{k=1}^{\infty} \left(\int_k^{k+1} \frac{1}{2025^{\lfloor x \rfloor}} dx \right) \\
 &= \sum_{k=1}^{\infty} \left(\int_k^{k+1} \frac{1}{2025^k} dx \right) \\
 &= \sum_{k=1}^{\infty} \frac{1}{2025^k} \\
 &= \frac{\frac{1}{2025}}{1 - \frac{1}{2025}} = \boxed{\frac{1}{2024}}
 \end{aligned}$$

Problem 9. $\int_4^5 \frac{2x+1}{(x-1)(x-2)(x-3)} dx$

Proposed by Sam Murray

Solution: $\boxed{\frac{23\ln(2) - 13\ln(3)}{2}}$

We note that the integrand will admit a partial fraction decomposition as follows.

$$\begin{aligned}
 \frac{2x+1}{(x-1)(x-2)(x-3)} &= \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3} \\
 2x+1 &= A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2) \\
 x=1 &\implies 3 = A(-1)(-2) \implies A = \frac{3}{2} \\
 x=2 &\implies 5 = B(1)(-1) \implies B = -5 \\
 x=3 &\implies 7 = C(1)(2) \implies C = \frac{7}{2}
 \end{aligned}$$

Therefore, we can integrate the partial fraction decomposition as follows.

$$\begin{aligned}
 \int_4^5 \frac{2x+1}{(x-1)(x-2)(x-3)} dx &= \int_4^5 \frac{3}{2(x-1)} - \frac{5}{x-2} + \frac{7}{2(x-3)} dx \\
 &= \left[\frac{3}{2} \ln(x-1) - 5 \ln(x-2) + \frac{7}{2} \ln(x-3) \right]_4^5 \\
 &= \left[\frac{3}{2} \ln(4) - 5 \ln(3) + \frac{7}{2} \ln(2) \right] - \left[\frac{3}{2} \ln(3) - 5 \ln(2) + \frac{7}{2} \ln(1) \right] \\
 &= \left[3 \ln(2) - 5 \ln(3) + \frac{7}{2} \ln(2) \right] - \left[\frac{3}{2} \ln(3) - 5 \ln(2) \right] \\
 &= \boxed{\frac{23\ln(2) - 13\ln(3)}{2}}
 \end{aligned}$$



Problem 10. $\int_0^1 x e^{x^2+e^{x^2}} dx$

Proposed by Sam Murray

Solution: $\boxed{\frac{e^e - e}{2}}$

One can recognize that taking the derivative of the following yields an expression similar to our integrand. We can then integrate both sides to obtain the desired result.

$$\begin{aligned}\frac{d}{dx} \left(e^{e^{x^2}} \right) &= e^{e^{x^2}} \cdot (e^{x^2})' \\ &= e^{e^{x^2}} \cdot e^{x^2} \cdot 2x \\ \int_0^1 2 \cdot x e^{x^2+e^{x^2}} dx &= \int_0^1 \frac{d}{dx} \left(e^{e^{x^2}} \right) dx \\ \int_0^1 x e^{x^2+e^{x^2}} dx &= \frac{1}{2} \cdot e^{e^{x^2}} \Big|_0^1 = \boxed{\frac{e^e - e}{2}}\end{aligned}$$

Problem 11. $\int_0^{\frac{3}{2}} \lfloor x + \lfloor 2x \rfloor \rfloor dx$

Proposed by Sam Murray

Solution: $\boxed{2}$

We can use the $\lfloor 2x \rfloor$ to split up the integral as follows.

$$\begin{aligned}\int_0^{\frac{3}{2}} \lfloor x + \lfloor 2x \rfloor \rfloor dx &= \int_0^{\frac{1}{2}} \lfloor x + \lfloor 2x \rfloor \rfloor dx + \int_{\frac{1}{2}}^1 \lfloor x + \lfloor 2x \rfloor \rfloor dx + \int_1^{\frac{3}{2}} \lfloor x + \lfloor 2x \rfloor \rfloor dx \\ &= \int_0^{\frac{1}{2}} \lfloor x \rfloor dx + \int_{\frac{1}{2}}^1 \lfloor x + 1 \rfloor dx + \int_1^{\frac{3}{2}} \lfloor x + 2 \rfloor dx \\ &= \int_{\frac{1}{2}}^1 1 dx + \int_1^{\frac{3}{2}} 3 dx = \frac{1}{2} + \frac{3}{2} = \boxed{2}\end{aligned}$$

Problem 12. $\int_{2025}^{2024} \ln(\{x\}) dx$

Proposed by Sam Murray

Solution: $\boxed{1}$



Recall that $\{x\} = x - \lfloor x \rfloor$, of which the latter term is constant on the interval as follows.

$$\begin{aligned} \int_{2025}^{2024} \ln(x - \lfloor x \rfloor) \, dx &= \int_{2025}^{2024} \ln(x - 2024) \, dx \\ &= \int_1^0 \ln(u) \, du \\ &= [u \ln(u) - u]_1^0 \\ &= \boxed{1} \end{aligned}$$

Problem 13. $\int_{98}^{99} x^{-\frac{2}{\log x}} \, dx$

Proposed by Sam Murray

Solution: $\boxed{e^{-2}}$

Upon letting $x = e^u$, we drastically simplify the integral and proceed normally.

$$\begin{aligned} \int_{98}^{99} x^{-\frac{2}{\log x}} \, dx &= \int_{\ln(98)}^{\ln(99)} (e^u)^{-\frac{2}{\log e^u}} (e^u \, du) \\ &= \int_{\ln(98)}^{\ln(99)} (e^u)^{-\frac{2}{u}} (e^u \, du) \\ &= \int_{\ln(98)}^{\ln(99)} \frac{e^u}{e^2} \, du \\ &= \frac{e^u}{e^2} \Big|_{\ln(98)}^{\ln(99)} = \frac{99 - 98}{e^2} = \boxed{e^{-2}} \end{aligned}$$

Problem 14. $\int_{-\infty}^{\infty} e^{-x^2+3x} \, dx$

Proposed by

Solution: $\boxed{e^{9/4} \sqrt{\pi}}$

We note that all densities integrate to 1, and the standard normal density allows us to make the following relation.

$$\begin{aligned} 1 &= \frac{1}{\sqrt{2\pi}} \cdot \int_{-\infty}^{\infty} e^{-x^2/2} \, dx \\ \sqrt{2\pi} &= \int_{-\infty}^{\infty} \exp(-x^2/2) \, dx \\ \sqrt{2\pi} &= \sqrt{2} \cdot \int_{-\infty}^{\infty} \exp(-u^2) \, du & (x = \sqrt{2}u, dx = \sqrt{2} \, du) \\ \sqrt{\pi} &= \int_{-\infty}^{\infty} \exp(-u^2) \, du \end{aligned}$$



Therefore, it suffices to complete the square in the desired integral since the above is translation invariant.

$$\begin{aligned}\int_{-\infty}^{\infty} \exp(-x^2 + 3x) \, dx &= \int_{-\infty}^{\infty} \exp\left(-x^2 + 3x - \frac{9}{4} + \frac{9}{4}\right) \, dx \\ &= \int_{-\infty}^{\infty} \exp\left(-\left(x - \frac{3}{2}\right)^2 + \frac{9}{4}\right) \, dx \\ &= e^{9/4} \cdot \int_{-\infty}^{\infty} \exp(-u^2) \, du \qquad \left(u = x - \frac{3}{2}, du = dx\right) \\ &= \boxed{e^{9/4} \cdot \sqrt{\pi}}\end{aligned}$$